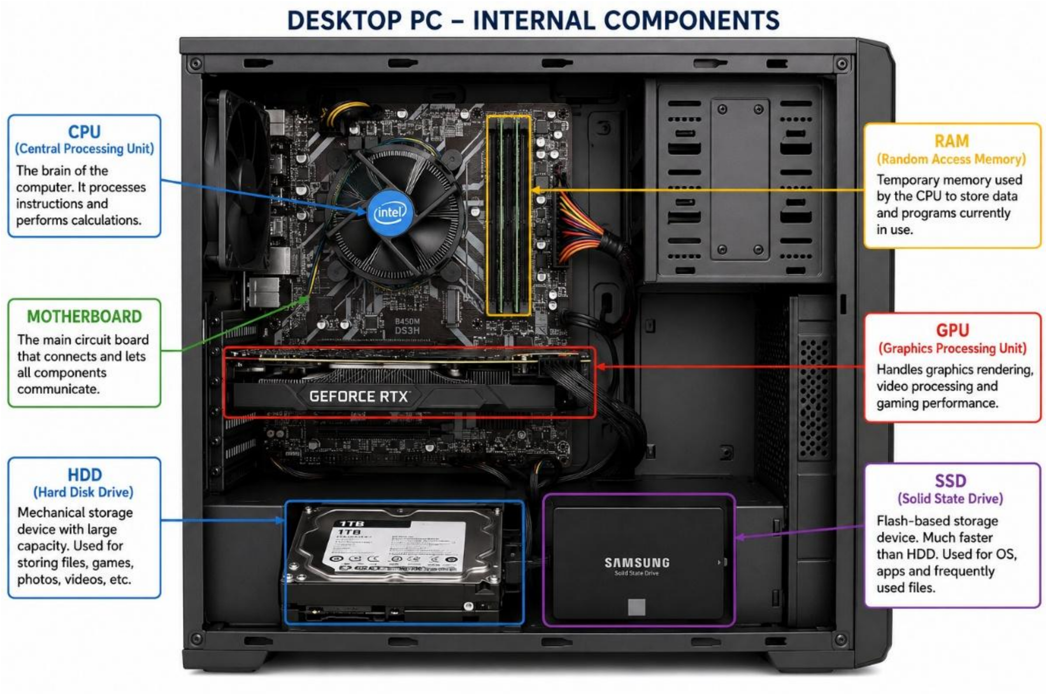
1. Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.

ANS:-



2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.Hint:Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).

Feature	**HDD (Hard Disk Drive)**	**SSD (Solid State Drive)**
Full Form	Hard Disk Drive	Solid State Drive
Speed	Slower	Much faster
Durability	Less durable; sensitive to shocks and drops due to moving parts	More durable; highly shock-resistant because it has no moving parts
Noise	Produces noise while spinning	Completely silent
Power Consumption	Higher	Lower (better battery life in laptops)
Heat Generation	Generates more heat	Generates less heat
Weight	Heavier	Lighter
Storage Capacity	Usually available in larger capacities at lower cost	Available in many capacities but costs more per GB
Cost	Cheaper	More expensive
Typical Use Cases	File storage, backups, media libraries, budget desktop PCs	Operating system, gaming PCs, laptops, professional workstations

3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

ANS:-

CPU (Central Processing Unit)

The **CPU** is the brain of the computer. It processes game instructions, controls player actions, manages physics, and coordinates communication between all hardware components. A faster CPU helps games like **PUBG** or **Free Fire** run more smoothly, especially during intense gameplay.

RAM (Random Access Memory)

RAM temporarily stores the game data and files that the CPU needs while the game is running. More RAM allows the game to load faster and switch between tasks without slowing down. Insufficient RAM can cause lag, stuttering, or longer loading times.

GPU (Graphics Processing Unit)

The **GPU** is responsible for rendering the game's graphics, including characters, maps, lighting, and visual effects. A powerful GPU delivers higher frame rates, better graphics quality, and smoother gameplay. In games like **PUBG** or **Free Fire**, the GPU greatly improves the overall visual experience and performance.

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

ANS:-

If I wanted to upgrade my PC to play the latest version of **FIFA** smoothly, I would focus on these components:

1. **Graphics Card (GPU):** I would upgrade the GPU because it is responsible for displaying the game's graphics. A better graphics card gives smoother gameplay, higher FPS, and better visual quality.

2. **Processor (CPU):** I would choose a faster CPU so the game can process player movements, AI, and match calculations more efficiently. This helps reduce lag during gameplay.

3. **RAM:** I would increase the RAM to at least **16 GB**. More RAM helps the game run smoothly and allows me to keep other applications open without slowing down the PC.

4. **SSD (Solid State Drive):** If my PC still uses an HDD, I would replace it with an SSD. This makes the game load

much faster and improves the overall speed of the computer.

5. **Power Supply (SMPS):** If I install a more powerful graphics card or processor, I would also upgrade the power supply to make sure the PC gets enough stable power.

6. **Cooling System:** I would improve the cooling with better fans or a CPU cooler. This keeps the PC from overheating during long gaming sessions and helps maintain good performance.

Overall, I think upgrading the **GPU, CPU, RAM, and SSD** would make the biggest difference. These upgrades would help FIFA run smoothly with better graphics, faster loading times, and fewer performance issues.